

## Question 1

```
CREATE TABLE department_ (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(30)  
);
```

```
CREATE TABLE employee_ (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50),  
    salary NUMERIC(10,2),  
    dept_id INT REFERENCES department_(dept_id)  
);
```

```
INSERT INTO department_ VALUES  
(1, 'IT'),  
(2, 'Sales'),  
(3, 'HR');
```

```
INSERT INTO employee_ VALUES  
(101, 'Aman', 90000, 1),  
(102, 'Neha', 70000, 1),  
(103, 'Raj', 50000, 1),  
(104, 'Priya', 80000, 2),  
(105, 'Karan', 60000, 2),  
(106, 'Riya', 40000, 2),  
(107, 'Mohit', 75000, 3),  
(108, 'Simran', 55000, 3);
```

## Solution:-

```
SELECT d.dept_name as Department, e.emp_name as Employee, e.salary
FROM employee_ e
JOIN department_ d
    ON e.dept_id = d.dept_id
WHERE e.salary > (
    SELECT AVG(e2.salary)
    FROM employee_ e2
    WHERE e2.dept_id = e.dept_id
)
AND e.dept_id IN (
    SELECT dept_id
    FROM employee_
    GROUP BY dept_id
    HAVING COUNT(*) >= 3
);
```

| Data Output   |                                      |                                    |                          |
|---------------|--------------------------------------|------------------------------------|--------------------------|
| Messages      |                                      |                                    |                          |
| Notifications |                                      |                                    |                          |
|               | department<br>character varying (30) | employee<br>character varying (50) | salary<br>numeric (10,2) |
| 1             | IT                                   | Aman                               | 90000.00                 |
| 2             | Sales                                | Priya                              | 80000.00                 |

## Question 2

```
CREATE TABLE bank_customer_ (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    balance NUMERIC(10,2)
```

```
);
```

```
CREATE TABLE customer_audit_ (  
    audit_id SERIAL PRIMARY KEY,  
    customer_id INT,  
    customer_name VARCHAR(50),  
    action VARCHAR(20),  
    action_time TIMESTAMP  
);
```

```
INSERT INTO bank_customer_ VALUES  
(1, 'Rahul', 50000),  
(2, 'Neha', 75000);
```

## **Solution:-**

```
CREATE OR REPLACE FUNCTION log_customer_changes()  
RETURNS TRIGGER  
LANGUAGE plpgsql  
AS $$  
BEGIN  
    IF TG_OP = 'INSERT' THEN  
        INSERT INTO customer_audit  
        (customer_id, customer_name, action, action_time)  
        VALUES  
        (NEW.customer_id, NEW.customer_name, 'ADDED', CURRENT_TIMESTAMP);  
  
    ELSIF TG_OP = 'DELETE' THEN  
        INSERT INTO customer_audit  
        (customer_id, customer_name, action, action_time)  
        VALUES
```

```

        (OLD.customer_id, OLD.customer_name, 'REMOVED', CURRENT_TIMESTAMP);
    END IF;

    RETURN NULL;
END;
$$;

```

```

CREATE TRIGGER customer_audit_trigger
AFTER INSERT OR DELETE
ON bank_customer
FOR EACH ROW
EXECUTE FUNCTION log_customer_changes();

```

```

INSERT INTO bank_customer VALUES
(3, 'Aman', 60000);

DELETE FROM bank_customer
WHERE customer_id = 1;

```

```

SELECT * FROM customer_audit;

```

| Data output Messages Notifications |                          |                        |                                         |                                  |                                            |  |
|------------------------------------|--------------------------|------------------------|-----------------------------------------|----------------------------------|--------------------------------------------|--|
|                                    | audit_id<br>[PK] integer | customer_id<br>integer | customer_name<br>character varying (50) | action<br>character varying (20) | action_time<br>timestamp without time zone |  |
| 1                                  | 1                        | 3                      | Aman                                    | ADDED                            | 2026-09-28 15:36:49.423727                 |  |
| 2                                  | 2                        | 1                      | Rahul                                   | REMOVED                          | 2026-09-28 15:36:49.423727                 |  |